Adaptive Memory Dynamics - Selective Consolidation Follow-Up

Exploratory software evidence · October 15, 2026 · not peer reviewed

Repository · Previous Stage Two findings · Full evidence bundle

Research question

Our original continuous parameter-decay mechanisms did not improve domain-incremental retention: persistent memory was more accurate on Task A and Task B on average. This follow-up tests a different, explicit mechanism: **consolidate Task A, then protect selected parameters from Task B updates**. Does selective protection improve the **stability/plasticity trade-off** relative to *uniform* protection and no protection?

The original Stage One and earlier Stage Two records are unchanged.

Implementation and controls

The data remain offline **scikit-learn 8 x 8 handwritten digits**, ten shared labels, and two synthetic Task B transformations: 90° rotation and seeded pixel permutation. No MNIST/CIFAR or human-memory claims. Five **new** paired seeds (1109, 1201, 1301, 1409, 1501) each share the same Task A checkpoint, A/B train/test split, and Task B minibatch order across eight configurations:

- Persistent: no anchoring.

- Uniform quadratic weight anchoring (penalties 0.1 and 0.5).

- Activity-weighted pathway anchoring (penalties 0.1 and 0.5).

- Squared *minibatch-gradient proxy* anchoring (penalties 0.1 and 0.5; **not an empirical Fisher/EWC benchmark**).

- Shuffled-activity importance (0.5): negative control testing whether pathway ranking matters.

All methods receive eight Task A epochs and eight Task B epochs, a 32-unit ReLU hidden layer and SGD at 0.08. Importance comes from Task A **training** samples only. This post-SGD quadratic step is a software prototype, not exact joint-loss optimization.

Observed held-out outcomes

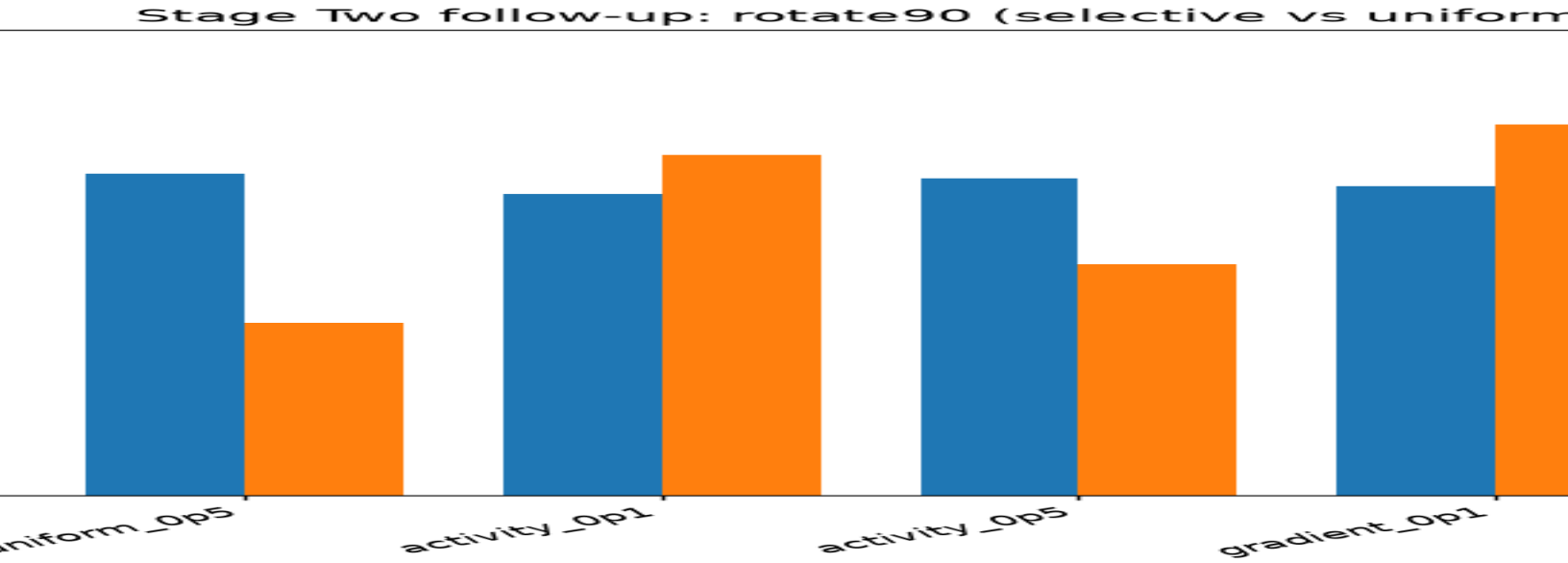
All values are mean proportions over **five paired seeds**; percent equivalents are included for readability. **Higher Task A after B** means better old-task retention; **higher Task B** means better current-task adaptation.

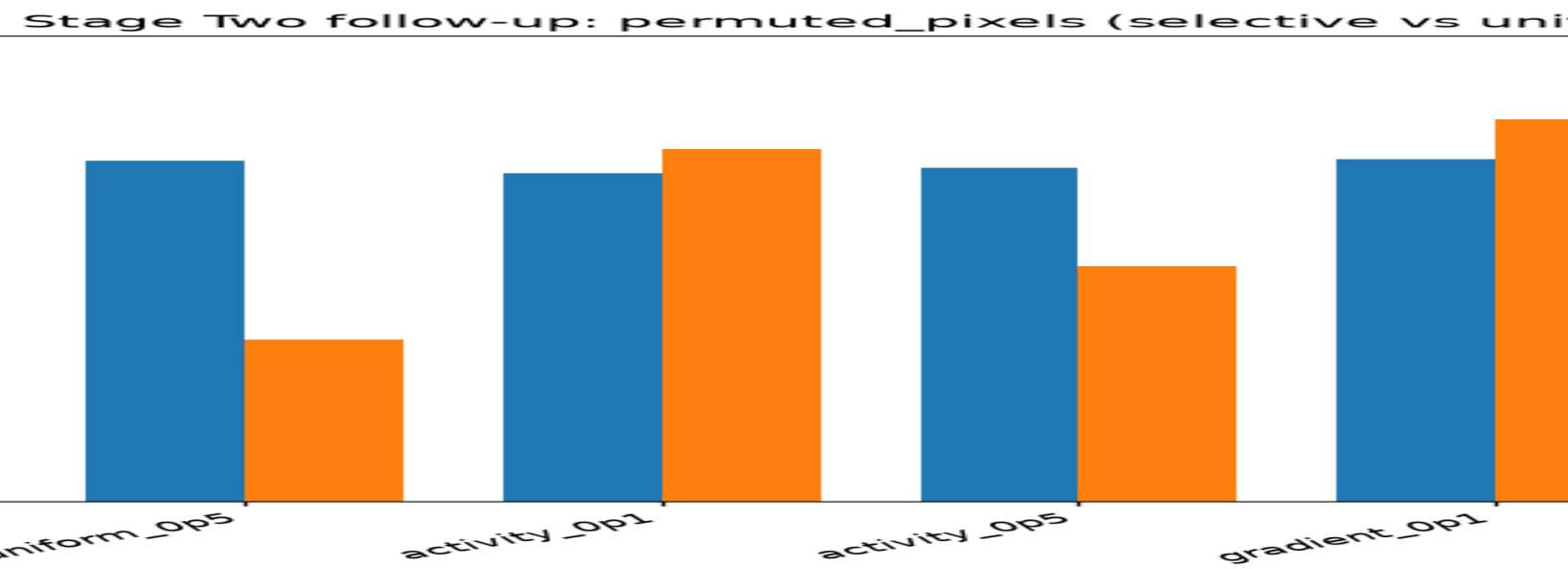
Treatment	Rotation A	Rotation B	Permutation A	Permutation B
Persistent	55.73%	**87.24%**	64.93%	**87.82%**
Uniform 0.1	69.02%	73.82%	71.47%	76.67%
Uniform 0.5	69.24%	37.16%	73.20%	34.76%
Activity 0.1	64.89%	73.24%	70.58%	75.78%
Activity 0.5	68.13%	49.73%	71.64%	50.49%
Gradient proxy 0.1	66.49%	79.69%	73.51%	82.09%

Treatment	Rotation A	Rotation B	Permutation A	Permutation B
Gradient proxy 0.5	**73.42%**	69.73%	**76.62%**	74.04%
Shuffled activity 0.5	65.91%	65.29%	69.73%	64.04%

The gradient-proxy 0.1 condition has the highest **unweighted mean of Task A and B accuracy** among the tested regularized alternatives: **73.09% for rotation** and **77.80% for permutation**, versus **71.49%** and **76.38%** for persistence. It therefore marginally exceeds persistence on this particular balanced measure but learns Task B less accurately. This is an exploratory trade-off, not independently validated superiority.

Key contrast: keeping Task A accuracy is possible without decay during training. More protection can reduce learning of B substantially. Gradient importance outperformed activity importance at the same penalty in this dataset; at strong penalty, shuffled importance also differs from the unshuffled mask, but neither wins both tasks.





Statistical interpretation

The linked `paired_effects.csv` contains 5-seed paired mean differences from persistent with descriptive 95% bootstrap intervals, unadjusted for the **many** treatments, endpoints, and two domains. This experiment is exploratory and its hyperparameters were designed after inspecting the original Stage Two results. Do **not** claim inferential statistical significance, generalized continual-learning superiority or a validated biological theory of selective forgetting. A future confirmatory study must predeclare one primary stability/plasticity metric, use a fresh held-out set of seeds, and compare strong standard baselines (such as replay and correctly implemented EWC).

Reproducibility

Archived manifest records software environment, source commit, seeds, treatments, and SHA-256 CSV digests. **80 paired condition runs** (5 seeds x 2 domains x 8 conditions), **1,280 epoch-level observations**, and **42 paired effect records** were generated by the successful CI run.

```bash
pip install -e ".[dev]"
ruff check .
pytest -q
python -m adaptive_memory.selection_followup --output results/selection-followup
```

The previous experiments and their research documents remain available without edits. This follow-up does not supersede the old negative results - it offers a different, testable explanation and direction.